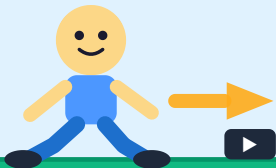


WALK

BEGINNER



MOTION

```
when [right arrow▼] key pressed
```

```
change x by (10)
```

```
when [left arrow▼] key pressed
```

```
change x by (-10)
```

// tip: flip costume

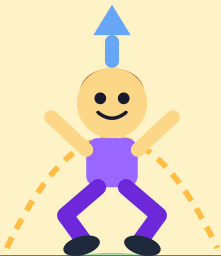
// for direction facing

**PRO TIP**

Use **change x** (not 'move') to prevent diagonal drift when pressing two keys at once.

JUMP

BEGINNER



SPACE

MOTION

```
when [space▼] key pressed
```

```
if <(on ground?) = true> then
```

```
  set [vy▼] to (-15)
```

```
end
```

// negative vy = upward

// gravity pulls back down

💡 **PRO TIP** Save velocity in a variable **vy**. Gravity adds to vy each frame — realistic feel!

GRAVITY

BEGINNER



v_y

8

VARIABLES

```
when  clicked
```

```
forever
```

```
change [vy▼] by (1)
```

```
change y by (vy)
```

```
if <touching [ground▼]>?
```

```
set [vy▼] to (0)
```

```
end
```

```
end
```

**PRO TIP**

Start vy at 0. Add 1 each frame. Cap vy at **15** so the character doesn't fall too fast.

COLLECT

BEGINNER

SCORE: 1



SENSING

```
when  clicked
```

```
forever
```

```
if <touching [Apple▼]?> then
```

```
change [score▼] by (1)
```

```
hide
```

```
end
```

```
end
```

**PRO TIP**

Hide the apple when collected instead of deleting — clones can't be deleted the same way!

RANDOM SPAWN

BEGINNER



A diagram on a dark blue background showing five yellow stars. One star in the center is highlighted with a larger, semi-transparent yellow circle. A dashed yellow line connects a small yellow dot on the top-left star to the center of the highlighted star. To the left of this line is a yellow rounded rectangle containing the text 'x: pick random'.

x: pick random

OPERATORS

```
when  clicked
```

```
forever
```

```
go to x: (pick random
```

```
-200 to 200)
```

```
y: (180)
```

```
wait (2) secs
```

```
end
```

 **PRO TIP** **pick random** works with negative numbers! Use -200 to 200 for full stage width.

CLONE

INTERMEDIATE

ORIGINAL

CLONE



CONTROL

```
when  clicked
```

```
forever
```

```
create clone of [myself▼]
```

```
wait (1) secs
```

```
end
```

— clone script —

```
when I start as a clone
```

```
show
```

**PRO TIP**

The **original** sprite can stay hidden — clones do all the work. Set visible only in clone script.

CLONE FALLS

INTERMEDIATE



CONTROL

```
when I start as a clone
```

```
go to x: (pick random
```

```
-200 to 200) y: (180)
```

```
forever
```

```
change y by (-5)
```

```
if <(y pos) < -180>
```

```
delete this clone
```

```
end
```

```
end
```

**PRO TIP**

Always **delete** clones when off-screen — too many clones slow Scratch to a crawl!

FIRE LASER

INTERMEDIATE



SPACE

CONTROL

when [space▼] key pressed

create clone of [Laser▼]

— *Laser sprite* —

when I start as a clone

show

repeat until <y > 180>

change y by (15)

end

delete this clone

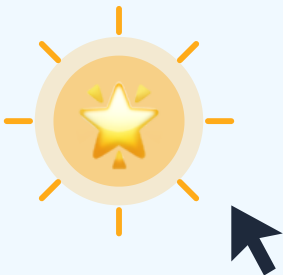
**PRO TIP**

Make the Laser sprite tiny (**2×12px** white rectangle). It moves faster than the ship.

MOUSE CLICK

INTERMEDIATE

+1



EVENTS

```
when this sprite clicked
```

```
change [score▼] by (1)
```

```
hide
```

```
wait (pick random 1 to 3) secs
```

```
go to x:(pick random
```

```
-200 to 200)
```

```
y:(pick random -150 to 150)
```

```
show
```

💡 **PRO TIP** **this sprite clicked** fires even if another script is running — great for quick reactions!



LIVES



broadcast: hit

```
when  clicked
```

```
set [lives▼] to (3)
```

— hit handler —

```
when I receive [hit▼]
```

```
change [lives▼] by (-1)
```

```
if <(lives) = 0> then
```

```
  broadcast [game over▼]
```

```
end
```

**PRO TIP**

Show lives as **costume changes** (heart sprites)
OR as a variable. Costume looks more fun!

SCORE TRACKER

INTERMEDIATE

SCORE

1250



+10



+10



+10

VARIABLES

```
when  🚩 clicked
```

```
  set [score▼] to (0)
```

— collect handler —

```
when I receive [collect▼]
```

```
  change [score▼] by (10)
```

// always reset on start

// use broadcast to add

**PRO TIP**

set score to 0 inside 'when flag clicked' so restarting always resets it.

TIMER

INTERMEDIATE



VARIABLES

```
when  clicked
```

```
set [time▼] to (30)
```

```
repeat until <(time) = 0>
```

```
  wait (1) secs
```

```
  change [time▼] by (-1)
```

```
end
```

```
broadcast [game over▼]
```

**PRO TIP**

Put the countdown in its **own script** so the game can still move while counting down.

DRAW BACKDROP

ADVANCED



BACKDROPS



LOOKS

PAINT EDITOR STEPS

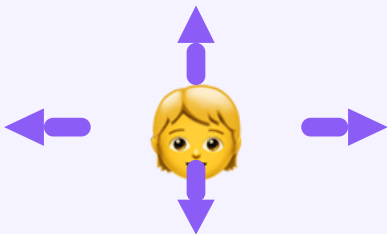
- 1 Click the **Backdrops** tab
- 2 Choose **Paint** (brush icon)
- 3 Draw maze walls in **black**
- 4 Fill background in **white**
- 5 Name it "**Level 1**"

**PRO TIP**

Use the **Fill tool** (paint bucket) to quickly color large areas. Zoom in for detail.

4-WAY MOVE

ADVANCED



WASD

MOTION

```
when  clicked
```

```
forever
```

```
if <key [up▼] pressed?>
```

```
change y by (5)
```

```
if <key [down▼] pressed?>
```

```
change y by (-5)
```

```
if <key [right▼] pressed?>
```

```
change x by (5)
```

```
if <key [left▼] pressed?>
```

```
change x by (-5)
```

```
end
```

**PRO TIP**

Sensing **key pressed?** inside forever lets **multiple keys** work simultaneously.



prev_x saved

```
when  clicked
```

```
forever
```

```
if <touching color [■]??>
```

```
go to x:(prev_x)
```

```
y:(prev_y)
```

```
end
```

```
set [prev_x▼] to (x pos)
```

```
set [prev_y▼] to (y pos)
```

```
end
```

**PRO TIP**

Save position **BEFORE** moving. If you hit a wall, teleport back — no getting stuck!

ROLLBACK

ADVANCED

SAVED
`prev_x,y`



MOTION

```
when  clicked
```

```
forever
```

```
set [prev_x▼] to (x pos)
```

```
set [prev_y▼] to (y pos)
```

```
move (5) steps
```

```
if <touching color []?>
```

```
go to x:(prev_x)
```

```
y:(prev_y)
```

```
end
```

```
end
```

**PRO TIP**

Save → Move → Check → Undo. A **universal pattern** for all collision detection!

FLAP PHYSICS

ADVANCED

$vy = -6$

↑ flap!



VARIABLES

```
when  clicked
```

```
forever
```

```
change [vy▼] by (0.4)
```

```
change y by (vy)
```

```
end
```

— flap —

```
when [space▼] key pressed
```

```
set [vy▼] to (-6)
```

**PRO TIP**

vy of **-6** on flap feels best. Gravity **+0.4** per frame. Tune these numbers!

SCROLL LEFT

ADVANCED



MOTION

— Pipe sprite —

```
when  clicked
```

```
forever
```

```
change x by (-4)
```

```
if <(x pos) < -260>
```

```
set x to (260)
```

```
end
```

```
end
```

**PRO TIP**

Move backdrop and obstacles **left**; keep the bird fixed in x. **Infinite scrolling!**

PROCEDURAL

ADVANCED



2 secs

2 secs



CONTROL

```
when 🚩 clicked
```

```
  forever
```

```
    create clone of [Pipe▼]
```

```
    wait (2) secs
```

```
  end
```

— clone script —

```
when I start as a clone
```

```
  set y to (pick random -80 to 80)
```

```
  repeat until <(x) < -260>
```

```
    change x by (-4)
```

```
  end
```

```
  delete this clone
```

**PRO TIP**

Randomize the **gap height** each clone. Players can't memorize a fixed pattern.

GAME OVER

ADVANCED

GAME OVER



broadcast sent!

CONTROL

```
when I receive [game over▼]
```

```
stop [other scripts▼]
```

```
switch backdrop to [Game Over▼]
```

```
wait (2) secs
```

```
stop [all▼]
```

**PRO TIP**

broadcast 'game over' from ANY sprite. One central script handles the ending **cleanly**.